



Need to Visualize Data Transcript

Good morning everyone. Welcome to BITS Pilani digital online programme. Today I will be taking you the contents of data visualisation and storytelling course.

We will be seeing introduction to data visualisation and specifically we will be looking at two topics. We will first understand what is the need to visualise data or to do a visualisation process and we will understand how visualisation helps us to communicate complex their information that we encounter in our day-to-day life. Okay, so the importance of visualisations in communicating our complex phenomenon with other audience, business people and how to do better decision-making.

We will understand these things today. Okay, so first of all let's answer the question why do we need to visualise data. Primary reason is 65% of us are visual learners which means we learn better when we see and perceive things.

Why? We retain about 80% of the information that we see as compared to 20% of the information that we read and also the brain takes 13 milliseconds to pro

cess an image compared to all other forms of information retrieval and perceiving. Visual information is processed very quickly retained at a higher rate therefore making visual imagery a potent source of information delivery. That's the reason why we need to visualise data.

So far good. So take a look at this slide. What does this slide is saying? It is summarising the voter turnout participation from the recent Lok Sabha elections.

As compared to 2019 there has been 7.4% increase in the eligible voters but as compared to 2019 elections the voter turnout ratio has fallen slightly from 67% to 66% and then you also have some information regarding the number of states or the states with higher voter turnout and the states with lower voter turnout. So the summary of the text conveys key pointers on the growth in the electoral base roughly close to 1 billion people are now eligible to vote in India of which only 66% have voted in the recent Lok Sabha election. States Lakshadweep and Assam have seen higher turnout at 80% whereas states like Bihar and Uttar Pradesh have seen lesser turnout less than 60% and national average is around 66%.

Once we read this text the brain will take certain time to process and retrieve the information. If I ask you about the same information after some time it is more likely that you miss out some details even though you have read the text multiple times. Now consider presenting the same information alongside with the help of a visual.

We take the help of a famous chart or a most used chart called horizontal bar or bar chart. This bar chart if you see how is it constructed on the x-axis it has the voter turnout ratio on the y axis it has statewide state names and each horizontal bar represents the voter turnout ratio and the vertical blue line represents the national average at 66%. Immediately you can conclude the states with which are coloured in red are below the national average and the ones in green are above national average



with Lakshadweep having the highest voter turnout at 84% Bihar having lowest voter turnout around 56% and the national average being at 66%.

The same information when combined with text conveys information more potently and most importantly it can be retained faster processed faster and there are three main reasons that why do you have to visualise data apart from communicating complex information. First of all why do you are visualising data it helps us in making decisions and how it helps us to make decisions it enables us to decipher patterns and trends in a short span of time and also most importantly it helps us to put forth an argument or convey a particular sense of story to our audience. The father of data visualisation Edward Tufte has captured this in a very succinct manner saying that visualisation helps you convey your story with credibility.

-----End-----